ESN on the Human Connectome

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***Abstract*—ESNs offer outstanding performance and efficiency in handling time-varying data. Their behavior depends on the structure of the reservoir weight matrix. We focus on understanding how they work, analyzing different network topologies to build the reservoir, both artificial and biologically inspired. To evaluate the effect of connectivity strength, we systematically vary the scaling parameter J and assess the resulting neural activity and variance. Our analysis includes six datasets corresponding to increasing levels of brain parcellation granularity (83 to 1015 nodes), and investigates both averaged and non-averaged connectome matrices to isolate the influence of structural variability. Results show that averaged connectomes exhibit early activation and chaotic dynamics, while non-averaged connectomes maintain more stable behavior with lower variance. We further assess predictive performance using the Mackey-Glass time series benchmark, finding that connectome-based reservoirs demonstrate biologically plausible yet less optimal performance compared to randomized topologies. These findings highlight the importance of structural heterogeneity in stabilizing recurrent dynamics and offer insight into the potential and limitations of connectome-informed reservoir computing, but still leave hope for future improvements.**

I. INTRODUCTION

THE Echo-State Network on the Human Connectome project aims to develop a machine learning framework—specifically, an Echo-State Network (ESN)—to investigate how the structural and dynamic properties of the human brain's connectome influence a network's capacity to learn and generalize across tasks. Since its inception, the project has achieved several key milestones. Notably, reservoirs have been constructed using various network topologies, and their performance has been systematically benchmarked. Furthermore, significant progress has been made in analyzing the connectivity patterns of the human brain. The datasets being used are from BrainGraph.org [1]

II. NOTATION

- $\rho \rightarrow$ (which is the largest absolute eigenvalue) denotes the spectral radius, which governs whether the ESN remains in the echo state property regime.
- $\alpha \rightarrow$ denotes the leak rate, which determines the timescale of reservoir dynamics. Lower values ($\alpha \ll 1$) encourage slow integration of past inputs, effectively extending memory, while higher values (α close to 1) lead to more reactive but shorter-lived dynamics.

$$\begin{aligned}\mathbf{x}(t+1) = (1-\alpha)\,\mathbf{x}(t) \\ + \alpha \cdot \tanh\left(\mathbf{W}_{\text{res}}, \mathbf{x}(t)\right. \\ + \mathbf{W}_{\text{in}}\,\mathbf{u}(t) + \mathbf{b}\end{aligned} \tag{1}$$

Where:

1) x(t) is the reservoir state at time t,
2) $\mathbf{W}_{\text{res}}$ is the recurrent (reservoir) weight matrix, and it controls the echo state property, VII-A
3) $\mathbf{W}_{\text{in}}$ is the input weight matrix,
4) u(t) is the input at time t,
5) b is the bias term,
6) α is the leak rate.

- leakage $\rightarrow$ enables the Echo-State Network to balance short-term responsiveness with long-term memory. By adjusting α, the network can be tuned to better handle tasks with varying temporal dependencies.

  Low leakage (small α):
  1) the network state updates slowly
  2) emphasizes past states
  3) suitable for tasks needing long-term memory

  High leakage (α close to 1):
  1) state rapidly reflects new inputs
  2) suitable for tasks needing fast reaction or short memory

$$\mathbf{x}(t+1) = (1-\alpha)\,\mathbf{x}(t) + \alpha \cdot f\big(\mathbf{W}_{\text{res}}\,\mathbf{x}(t) + \mathbf{W}_{\text{in}}\,\mathbf{u}(t) + \mathbf{b}\big) \tag{2}$$

Where:

1) f is typically a nonlinear activation function

III. ESN

Echo State Networks represent a distinct category within recurrent neural architectures, developed specifically to handle time-varying data patterns. In contrast to conventional RNNs that demand intricate adjustment of all connection weights, ESNs operate on a fundamentally different principle: they utilize a fixed, randomly initialized internal network—the reservoir. This structure transforms inputs into rich dynamical representations, effectively capturing temporal relationships and nonlinearities while significantly reducing computational demands. Training focuses exclusively on optimizing output connections, typically through straightforward regression methods.

At the core of ESN functionality lies the echo state property (ESP) VII-A, a mathematical guarantee that historical input influences gradually diminish over time, ensuring computational stability. Network behavior depends critically on two parameters: the spectral radius (ρ) of the reservoir's weight matrix, which balances stability against memory capacity, and the leak rate (α), which regulates how quickly or slowly the network integrates incoming information.

The elegant compromise between computational simplicity and representational capability has established ESNs as valuable tools across numerous fields. Researchers and practitioners have successfully deployed them for:

- Financial and meteorological forecasting
- Modeling and controlling complex systems
- Extracting patterns from noisy signals
- Simulating neural processes and brain-like computations

ESNs excel in computational efficiency while delivering impressive results on sequential data tasks, making them ideal for studying intricate, structured domains like the human brain's neural architecture.

Our project employs ESNs to explore how various network architectures—particularly those inspired by actual human connectome data—influence a reservoir's temporal information processing capabilities. This interdisciplinary approach creates a bridge between computational methods and neuroscience, examining whether networks modeled after biological structures might improve learning efficiency, memory performance, and dynamic complexity in recurrent neural systems.

IV. Types of Reservoirs

The structure of the reservoir weight matrix $\mathbf{W} \in \mathbb{R}^{N \times N}$ affects the dynamics of the ESN and its memory capacity. We tested 3 different reservoirs: Erdős-Rényi, fixed matrix values and Gaussian.

A. *Erdős-Rényi*

Weights are initialized randomly. Usually with the formula

$$W_{ij} = \begin{cases} \text{Uniform}(-a, a) & \text{with probability } p \\ 0 & \text{otherwise} \end{cases} \tag{3}$$

Where:

- a is the scaling factor of reservoir and
- p is the probability of connection between nodes, controlling sparsity.

The reservoir weight matrix is then rescaled to set its spectral radius $\rho(W)$, to a desired predefined value.

B. *Fixed matrix values*

Rather than initializing the reservoir weight matrix $\mathbf{W}_{\text{res}}$ with randomly generated values, we adopt a deterministic structure in which each neuron is assigned a fixed weight of 0.5 to its subsequent neuron and -0.5 to its preceding neuron. This structured configuration is designed to eliminate the variability introduced by stochastic initialization, enabling a more controlled and interpretable evaluation of the network's behavior. In particular, it facilitates a clearer assessment of the effects of external parameters such as:

- varying ρ,
- varying the number of neurons in the reservoir,
- varying the leak rate α.

C. *Gaussian*

Weights are drawn from a Gaussian distribution.

$$W_{ij} = \mathcal{N}(0, \sigma^2) \tag{4}$$

V. Connectome Data

A. *The Human Connectome Project*

The datasets we work with represent structural brain connectivity maps created from the Human Connecome Project (HCP) data. The HCP used diffusion MRI (dMRI) to map white matter connections in the human brain. White matter constitutes approximately half of the total brain volume, and it has a crucial role in regulating neural activity by connecting different areas, thus influencing learning and memory. It is also important to note that dMRI is a non-invasive technique to map the brain in vivo. In particular, dMRI tracks the diffusion of water molecules moving freely along axon bundles (also called white matter tracts, they form the wiring of the brain). In the HCP they worked with high spatial resolution across multiple diffusion-weighting shells, measuring 90 different (non-colinear) directions. Shells can be seen as levels of sensitivity to slower or more restricted diffusion. So water diffusion patterns captured by the data reveal how efficiently different brain regions can communicate.

B. *The braingraph.org database*

The braingraph.org team [1] processed the raw HCP diffusion data through a number of computational steps. For a detailed description of their methods we recommend reading their original publication, here we focus on the relevant characteristics of the main networks that we work with:

1) 86 nodes set, 1064 brains, 1 000 000 streamlines, 10x repeated&averaged
2) 129 nodes set, 1064 brains, 1 000 000 streamlines, 10x repeated&averaged
3) 234 nodes set, 1064 brains, 1 000 000 streamlines, 10x repeated&averaged
4) 463 nodes set, 1064 brains, 1 000 000 streamlines, 10x repeated&averaged
5) 1015 nodes set, 1064 brains, 1 000 000 streamlines, 10x repeated&averaged
6) One single averaged consensus connectome with 1015 nodes

Here:

- The number of nodes represents different resolutions i.e. levels of brain parcellation granularity. For example, the 234-node set divides the brain into 234 anatomically distinct regions, which in particular is based on the Lausanne atlas.

- Data from 1064 human subjects was processed.
- Streamlines are the computational reconstructions of fiber pathways, and in this case the maximum number of fiber pathways tracked was 1,000,000.
- The tracking algorithm was run 10 times and results were averaged to make them more reliable.

In the braingraphs the spatial trajectories are ignored and the focus is on the presence/absence of connections between the nodes, also called regions of interest (ROIs). The edges of the graphs are defined based on the physical properties of the neural fibers connecting the corresponding ROIs. In fact, there are various attributes of edges that can be used in our analysis, based on how we want to define the weights of the networks. We will later compare these, but for now, this is a list of the attributes of the graphs:

- number of fibers: count of reconstructed streamlines between ROIs. More fibers = stronger structural connection
- FA mean: Mean Fractional Anisotropy of fibers, represents white matter integrity (0 = isotropic diffusion of fibers, 1 = perfectly aligned fibers)
- fiber length mean: Average streamline length (mm)
- FA std: represents the variability in white matter organisation along the pathway
- fiber length std: represents the consistency of reconstructed pathway lengths

Clearly, the data we work with is both biologically robust and computationally tractable.

VI. Connectome and Reference networks in function of J

We investigated how the reference networks and connectome data behave as the coupling strength parameter, J, varies between different regimes. The critical value J marks a phase transition point between ordered and chaotic dynamics, a region often referred to as the "edge of chaos." Operating near this criticality is hypothesized to enhance computational capacity, memory retention, and generalization in recurrent networks. This occurs since at the "edge of chaos" the network is neither too stable, which limits memory, nor too chaotic, which hinders learning.

A. Reference Networks

We investigate the influence of network topology on the dynamic behavior of recurrent neural networks by analyzing three distinct connectivity structures: **Fully Connected**, **Gaussian Random**, and **Erdős-Rényi** networks. Each network comprises the same number of neurons and is governed by the same update rule, but differs in how the recurrent weight matrix $\mathbf{W}_{\mathrm{res}}$ is initialized. The Fully Connected network includes nonzero connections between all neuron pairs; the Gaussian Random network has weights drawn independently from a normal distribution with zero mean and scaled variance; and the Erdős-Rényi network uses a binomially distributed sparsity pattern, where each potential connection exists with fixed probability. To systematically study the effect of connection strength, we scale the weight matrices by a parameter J and measure resulting changes in network activity and variance. This comparative framework allows us to identify critical transitions and assess the robustness of each topology under varying dynamical regimes.

1) Fully Connected network:
   - mean:
     - At J $<$ 4, the network remains entirely inactive (mean activity approximately 0).
     - At J $\sim$ 4, a sharp transition occurs, neurons start to activate and mean activity rises.
     - At J $>$ 4, activity increases and gradually saturates.
     - This behaviour is likely due to nonlinear activation becoming active only when recurrent inputs exceed a certain threshold, before that the network remains in a linear regime.
   - variance:
     - At J<6,5, the variance is very low, suggesting that all neurons behave similarly or are all near zero (which is the case, as seen in the mean. The network is too weakly connected to propagate signals)
     - At J $\sim$ 6,5, the variance has a huge spike, indicating a chaotic behaviour
     - At J $>$ 6,5 the network has entered an unstable regime.

   Hence, the critical J value is at 4.

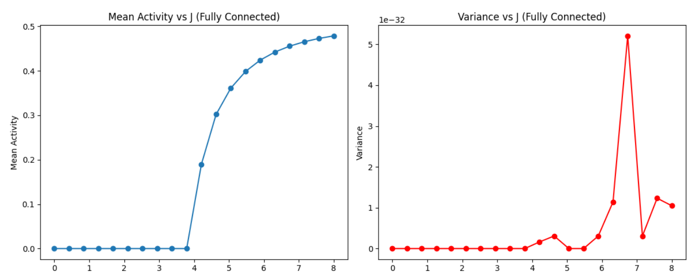

Fig. 1. fully connected network

2) Erdős-Rényi network:
   - mean:
     - At J<1.5, the network remains entirely inactive (mean activity roughly 0).
     - At J $\sim$ 1.5, a sharp transition occurs, neurons start to activate.
     - At J>1.5, activity saturates rapidly, indicating a phase transition from a quiescent state to a regime of active dynamics
   - variance:
     - At J<1,5, the variance is very low, suggesting that either all neurons behave similarly or they are all near zero (which is the case, as seen in the mean)

- At J $\sim$ 1,5, the variance has a huge spike, indicating a chaotic behaviour
- At J $>$ 1,5 the network's activity has saturated despite being very high, therefore variance decreases.

Hence, the critical J value is at 1.5

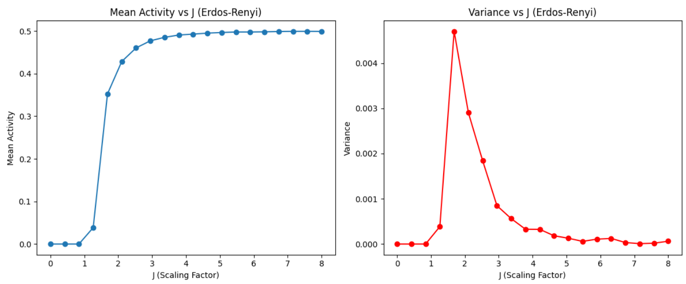

Fig. 2. Erdős-Rényi network

3) Gaussian:
   - mean:
     - At J<4, the network remains entirely inactive (mean activity approximately 0).
     - At J $>$ 4, the mean starts to oscillate drastically, suggesting a phase of chaotic behaviour
   - variance:
     - At J<4, the variance is very low, suggesting that either all neurons behave similarly or they are all near zero (which is the case, as seen in the mean)
     - At J $\sim$ 4, the variance has an important spike, indicating a chaotic behaviour
     - At J $>$ 4 the variance keeps increasing, indicating a phase of chaotic behaviour

   Hence, the critical J value is at 4

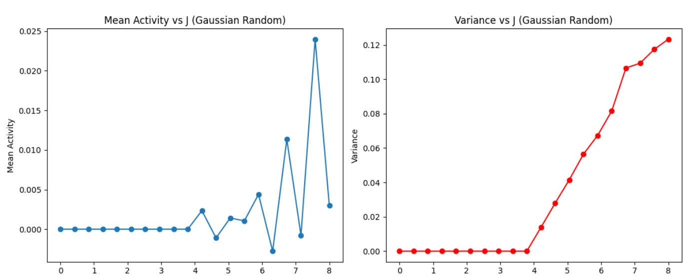

Fig. 3. gaussian network

B. Connectome Data

We analyze the dynamics of the recurrent neural networks from V-B

For each network, we examine how the dynamic behavior evolves as a function of the recurrent weight scaling factor J, which modulates the strength of internal connectivity. To investigate the effect of input structure and network heterogeneity, we conduct our analysis under two distinct conditions: in the first, we average all connectome matrices into a single effective connectivity matrix prior to simulation; in the second, we retain the individual connectome matrices without averaging. The averaged case provides insight into the macroscopic behavior representative of the entire population, reducing the influence of individual variability. In contrast, the non-averaged approach preserves network-specific structural differences, allowing us to explore the role of individual connectome topology in shaping network dynamics.

- Averaged connectomes:
  In the case of the averaged connectomes, we observe that the neurons tend to activate almost immediately as the scaling parameter J increases, even for relatively low values. This early onset of activation suggests that the averaging process amplifies the overall connectivity, effectively reducing the sparsity and homogenizing the weight distribution. As a result, the network operates in a high-gain regime, which pushes the system beyond the critical threshold required for complex dynamics. Consequently, the network exhibits highly chaotic behavior, characterized by large fluctuations in activity and a loss of stable or regular patterns. This highlights how averaging can obscure fine-grained structural features of individual connectomes and lead to artificially induced instability in the network's dynamics.

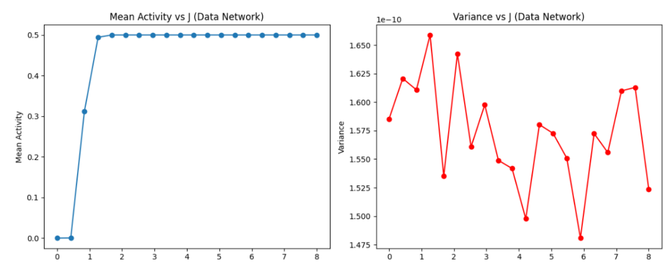

Fig. 4. averaged connectome

- Non-averaged connectomes:
  In contrast, the non-averaged connectomes exhibit more consistent and structured dynamics across the different network sizes. All networks show an initial sharp spike in activity as the scaling parameter J increases, often approaching saturation levels. However, despite this high initial activation, the variance of neuronal activity remains relatively low throughout. This suggests that, although the networks enter an active regime quickly, the individual neuron responses remain relatively uniform and temporally stable. The preservation of each connectome's unique topology likely contributes to this effect by maintaining biologically relevant heterogeneity and connectivity patterns, which act to constrain the network's overall variability and prevent the emergence of chaotic behavior. This highlights the importance of structural diversity in stabilizing neural dynamics.

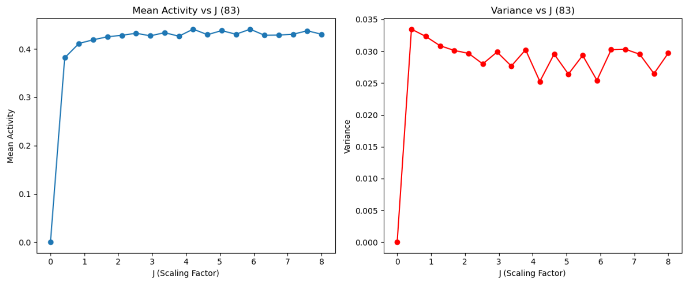

Fig. 5. 83 nodes

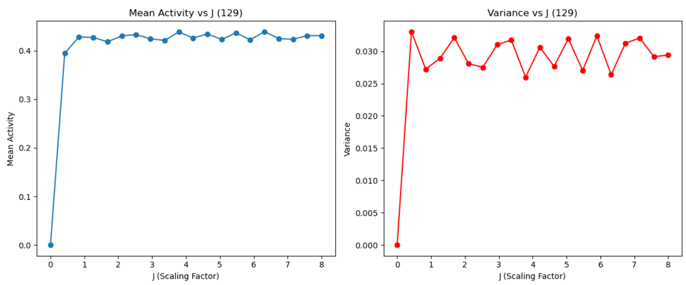

Fig. 6. 129 nodes

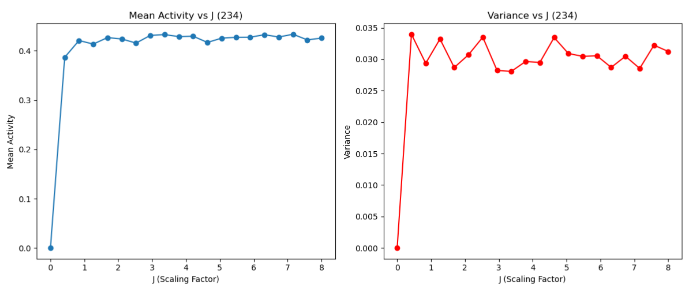

Fig. 7. 234 nodes

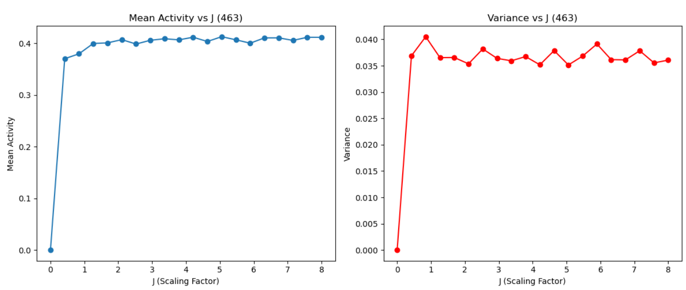

Fig. 8. 463 nodes

VII. Implementation of Gaussian ESN

A. Mathematical Backgrounds of ESN

Just like the architecture of RNN, there are three parts of the Echo State Networks: input layer, reservoir, and output layer. These components are connected with different weight matrices: the input weight matrix $\mathbf{W}_{\text{in}} \in \mathbb{R}^{N \times K}$, the reservoir weight matrix $\mathbf{W} \in \mathbb{R}^{N \times N}$, the feedback weight matrix $\mathbf{W}_{\text{fb}} \in \mathbb{R}^{N \times L}$, and the output weight matrix $\mathbf{W}_{\text{out}} \in \mathbb{R}^{L \times (N+K)}$. Here, K denotes the dimension of the

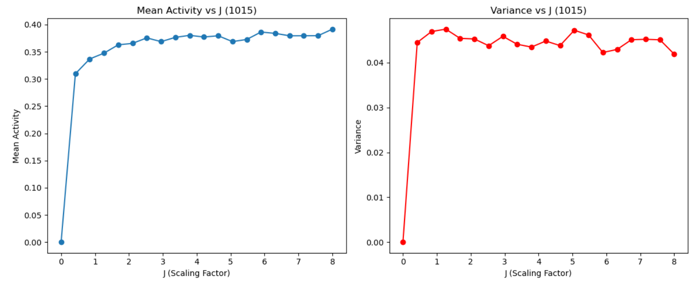

Fig. 9. 1015 nodes

input data, N is the number of reservoir neurons, and L is the dimension of the output data. [14]

At discrete time step n, the intermediate neuron update $\tilde{x}(n) \in \mathbb{R}^N$ is computed as a nonlinear transformation of the current input and the previous reservoir state:

$$\tilde{\mathbf{x}}(n) = \tanh\left(\mathbf{W}^{\text{in}}[1; \mathbf{u}(n)] + \mathbf{W}\mathbf{x}(n-1)\right)$$

where $u(n) \in \mathbb{R}^K$ is the input vector at step n and $x(n-1) \in \mathbb{R}^N$ is the reservoir state at the preceding time step. The concatenation term $[1; u(n)]$ includes a bias value. Other functions instead of $\tanh(\cdot)$ can be used, but it is the most common choice. [15]

Subsequently the reservoir state is updated using a leaky integrator architecture

$$\mathbf{x}(n) = (1-\alpha)\mathbf{x}(n-1) + \alpha\tilde{\mathbf{x}}(n),$$

, where $a \in (0, 1]$ is the leaking rate.

The linear readout layer in an Echo State Network (ESN) is defined by the equation

$$\mathbf{y}(n) = \mathbf{W}^{\text{out}}[1; \mathbf{u}(n); \mathbf{x}(n)],$$

where $\mathbf{y}(n) \in \mathbb{R}^{N_y}$ represents the network output at time step n, $\mathbf{W}^{\text{out}} \in \mathbb{R}^{N_y \times (1+N_u+N_x)}$ is the output weight matrix, and the notation $[\cdot; \cdot; \cdot]$ denotes concatenation of vectors (bias term, input vector $\mathbf{u}(n)$, and reservoir state $\mathbf{x}(n)$).

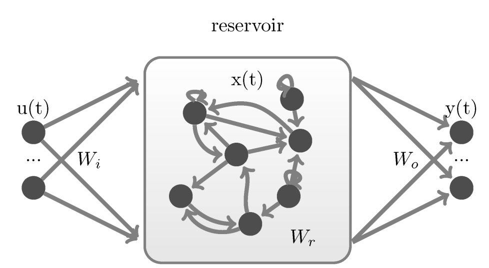

Fig. 10. Structure of the Echo State Network (ESN).[4]

B. Reservoir Initialization

The input weight matrix $\mathbf{W}_{\text{in}} \in \mathbb{R}^{N \times K}$ is initialized with values distributed in the range $[-1, 1]$.

The reservoir weight matrix $\mathbf{W} \in \mathbb{R}^{N \times N}$ is initialized with values from a Gaussian distribution with zero mean and a variance of J^2/N, where J determines the scale of the weights. Afterwards, we removed self-connections of the neurons to have a stable network dynamic.

In order to ensure Echo State Property, the spectral radius $\rho(W)$ is scaled to 1.25, providing the regulation of the stability and the memory capacity of the reservoir. In our model we used a spectral radius of 1.25

C. Updating Reservoir

At each discrete time step n, the reservoir state vector $x(n) \in \mathbb{R}^N$ is updated using a leaky-integrator model:

$$\mathbf{x}(n) = (1-\alpha)\mathbf{x}(n-1) \\ + a \tanh\left(\mathbf{W}^{\text{in}}[1; \mathbf{u}(n)] + \mathbf{W}\mathbf{x}(n-1)\right)$$

The leaky integration property allows ESNs to adjust the memory of reservoir neurons, which increases their adaptability to activities with different temporal characteristics.[3]

During training, random initial conditions could have an impact on the system, which could distort the learned dynamics of the system. Thus, we introduce a washout phase which simply allows us to ignore a number of steps at the beginning and lets us work with meaningful and stable dynamics. Once the washout phase finishes, for each discrete time step n, the internal state of the reservoir $x(n)$ is recorded. In our model, we used a washout phase of 100 discrete time points.

In order to train the output weight matrix $\mathbf{W}_{\text{out}} \in \mathbb{R}^{L\times(N+K)}$, an augmented state matrix X is created:

$$X\,(1; u(n), x(n))$$

, which consist of a constant bias term, 1 in our case, the input $u(n)$ and the reservoir state $x(n)$. This matrix is used consequently to train the readout layer by fitting a linear model.

D. Output Weight Training

The output of the system $y(n) \in \mathbb{R}^L$, is calculated by:

$$y(n) = W_{\text{out}}\,X(n)$$

, where $W_{\text{out}} \in \mathbb{R}^{L\times(1+K+N)}$ is the output weight matrix.

In order to train the output weight matrix, we use ridge regression. The goal is to find the W_{out} that minimizes the squared error between the model's outputs and the target outputs and while balancing the weights in order to avoid overfitting, by solving the equation:

$$W_{\text{out}} = \arg\min_{W} \|WX - Y_t\|_2^2 + \lambda\,\|W\|_2^2$$

, where λ is the penalty for large weights.

E. Prediction

At each discrete time step, the reservoir state is updated using the previous output:

$$\mathbf{x}(n) = (1-\alpha)\mathbf{x}(n-1) \\ + a \tanh\left(\mathbf{W}^{\text{in}}[1; \mathbf{u}(n)] + \mathbf{W}\mathbf{x}(n-1)\right)$$

, and the output is computed:

$$y(n) = W_{\text{out}}\,X(n)$$

which allows the ESNs to create predictions by solely training the parameters of output weights matrix W_{out}.[5]

F. Results of the Application

We measured the performance of the model on predicting the Mackey-Glass time series data. To asses the performance of the model we used mean squared error (MSE):

$$\text{MSE} = \frac{1}{N}\sum_{i=1}^{N}(\text{target}_i - \text{prediction}_i)^2$$

, over a evaluation window of 2000 data points.

We found that our model was able to predict the points with a MSE of 0.00002116609587399093, with 200 reservoir neurons, a relatively small number.

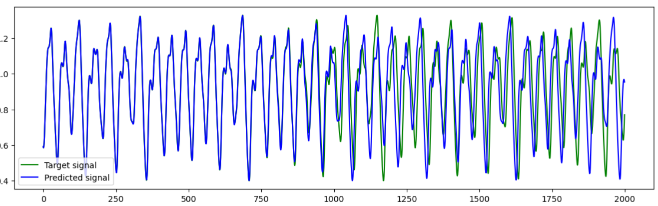

Fig. 11. Predicted and Target Values for the Mackey-Glass Time Series

VIII. Comparing Connectome-based networks with reference networks

A. Reference networks

First, we built our reference networks (fully-connected, Erdős-Rényi, Gaussian), making sure that they operate at the critical connectivity strength (J = 4, 1.5, 4, respectively) and that they satisfy the Echo State Property (ESP), for which we scaled them to have the spectral radius close to 1.25. This is to ensure that the initial weights of the connectivity matrix don't drive the predictions, but the outputs are rather influenced by the more recent inputs. All three networks were initiated with N = 234 nodes, because we later compare them with 234-node connectome-based networks.

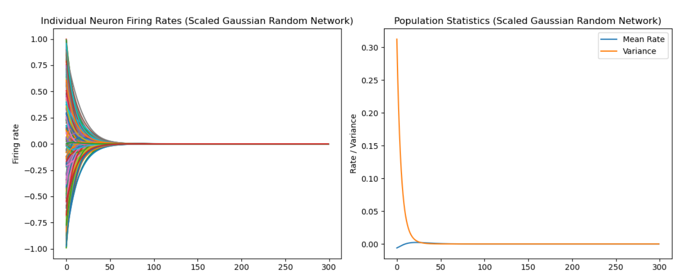

Fig. 12. This is the evolution of the Gaussian network, but the other two look very similar. They all have the 234 nodes initially uniformly distributed over the [-1,1] interval, to then quickly converge, which shows that they indeed satisfy the ESP.

We first employ graph-theoretic measures to better understand the structure of the networks, namely the clustering coefficients and the average path lengths. The clustering coefficient measures how much nodes in a graph tend to cluster together, specifically the likelihood that two neighbours of a node are neighbouring themselves (so it essentially counts the triangles in the network). The

The clustering coefficients (CC) of each of the three networks are:

- CC (Fully Connected): 1.0
- CC (Erdős-Rényi): 0.194
- CC (Gaussian Random): 0.089

And here we report the average path lengths (APL):

- APL (Fully Connected): 1.0
- APL (Erdős-Rényi): 1.808
- APL (Gaussian Random): 2.062

For the FC network perfect clustering and reachability are expected, since every node connects to all others. The ER network shows higher clustering and shorter paths than the Gaussian network, which indicates that it has more triangular structures, and then also more direct routes between distant nodes.

We then tested all three networks using Mackey-Glass time series data, which is a common benchmark for testing prediction algorithms, among which reservoir computing systems.

We measured the performance of each network by Memory Capacity (MC) and R^2. Memory Capacity = 1 - NRMSE (Normalised Root Mean Square Error) quantifies how well the network can reproduce pas inputs after a delay. $R^2 = 1-(SS_{res}/SS_{tot})$ measures proportion of variance explained by the model, and contrary to Memory Capacity it is scale-independent, because it doesn't work with absolute errors. These are the results we got:

- FC: MC: 0.6804 R^2: 0.8978
- ER: MC: 0.8973 R^2: 0.9895
- GR: MC: 0.8963 R^2: 0.9893

The fully connected network has moderate memory retention, it is able to capture most patterns, but it struggles with accuracy. This is likely because of its homogeneous connectivity, which limits its ability to deal with complex patterns. Erdős-Rényi and Gaussian random both have excellent memory and near-perfect pattern matching, thanks to their rich dynamics provided by randomised sparse connections.

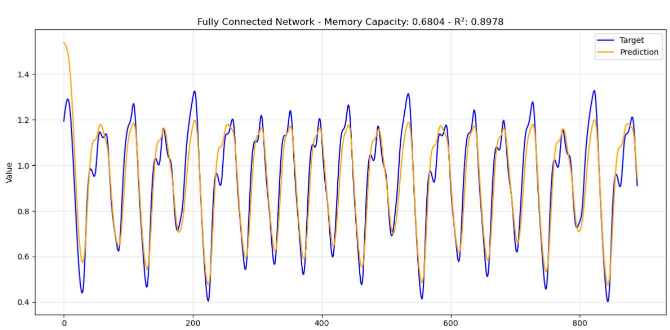

Fig. 13. Predictive performance of Fully-connected network.

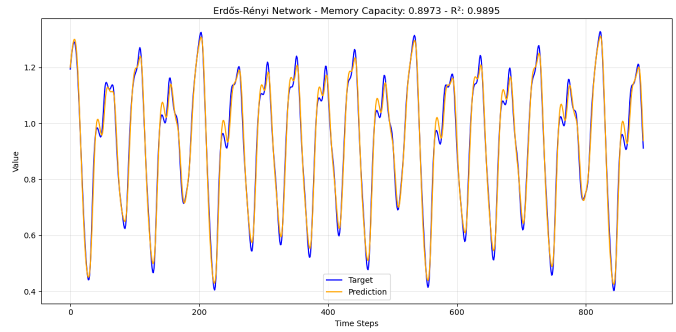

Fig. 14. Predictive performance of Erdős-Rényi network.

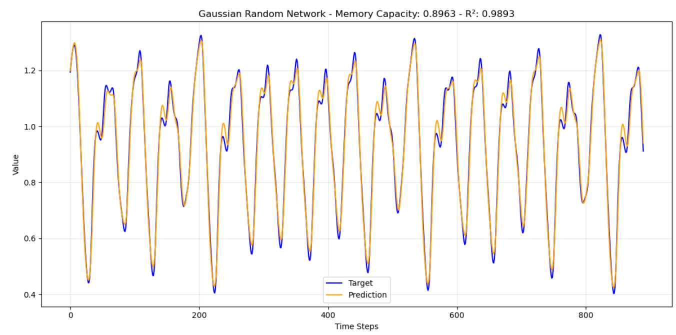

Fig. 15. Predictive performance of Gaussian random network.

B. Connectome-based networks

We now take the 234-node connectome-based network.

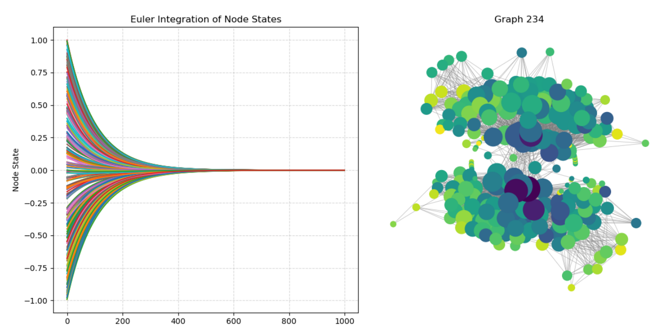

Fig. 16. 234-node connectome network based on a single brain. It has been scaled similarly to the reference networks, so that it satisfies the ESP.

The reason we chose this network to include here is for the balance it provides between a detailed representation of the brain and computational tractability. The network is Lyapunov-stable and robust to noise.

The clustering coefficient here was CC = 0.2751, and APL = 2.5731 the average path length. This is biologically consistent, with high clustering and short connection paths in the brain.

When tested with Mackey-Glass time series data it had the following performance:

- MC: 0.8125 R^2: 0.9648

This is lower than what we had for the Erdős-Rényi and the Gaussian random networks. This could be due to a number of reasons, here we speculate. It could be because the connectome-based connectivity is less random and more structured than both the Erdős-Rényi and the Gaussian networks. Due to high clustering connectomes are also more efficient at local processing and integration compared to global activity.

We are still exploring how to improve the performance of our connectome-based network and for which tasks it could be more efficient than the reference networks. Increasing resolution (using more nodes), introducing some "artificial" randomness, and trying various weight initialisations are all possible solutions we look at. We predict that a combination of these techniques will result in a better performance, even higher than the Erdős-Rényi network in this case.

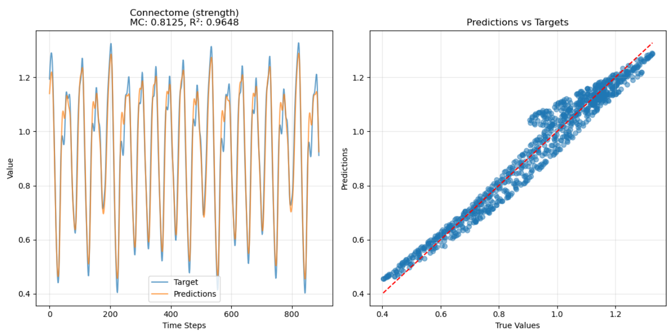

Fig. 17. Strength-weighted 234-node connectome network performance on synthetic time series data.

IX. Application

Once a robust connectome-based model with optimal performance is developed, it can be employed for a variety of tasks, either directly, or used as a benchmark (the latter is because all connectome data we use comes from healthy brains).

For example, for early diagnosis of neurological diseases, such as Alzheimer's, where memory is critical. A connectome-based RC model trained on cognitive tasks (e.g., memory recall) would show reduced performance when initialized with Alzheimer's patient connectomes, even in pre-symptomatic stages.

There's also autism, which involves atypical connectivity in social brain networks (e.g., fronto-temporal). A connectome-based RC model trained on social cognition tasks (e.g., emotion recognition) would show divergent activation patterns in autistic vs. neurotypical reservoirs.

Another application could be epilepsy focus localisation. Epileptic networks show hypersynchronous hubs. A connectome-based RC model simulating seizure propagation could identify "critical nodes" where perturbations most disrupt dynamics.

These are novel approaches in applying machine learning models, and quite valid in this case given the biological foundations of the model. However, it is also promising for traditional machine learning tasks, such as speech and audio signal prediction, forecasting chaotic systems, or, switching to classification tasks, pattern recognition and brain state decoding.

X. References Section

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